

Addressing Inconsistent Labeling With Cross Image Matching for Scribble-Based Medical Image Segmentation

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Abstract—In recent years, there has been a notable surge in the adoption of weakly-supervised learning for medical image segmentation, utilizing scribble annotation as a means to potentially reduce annotation costs. However, the inherent characteristics of scribble labeling, marked by incompleteness, subjectivity, and a lack of standardization, introduce inconsistencies into the annotations. These inconsistencies become significant challenges for the network’s learning process, ultimately affecting the performance of segmentation. To address this challenge, we propose creating a reference set to guide pixel-level feature matching, constructed from class-specific tokens and pixel-level features extracted from variously images. Serving as a repository showcasing diverse pixel styles and classes, the reference set becomes the cornerstone for a pixel-level feature matching strategy. This strategy enables the effective comparison of unlabeled pixels, offering guidance, particularly in learning scenarios characterized by inconsistent and incomplete scribbles. The proposed strategy incorporates smoothing and regression techniques to align pixel-level features across different images. By leveraging the diversity of pixel sources, our matching approach enhances the network’s ability to learn consistent patterns from the reference set. This, in turn, mitigates the impact of inconsistent and incomplete labeling, resulting in improved segmentation outcomes. Extensive experiments conducted on three publicly

available datasets demonstrate the superiority of our approach over state-of-the-art methods in terms of segmentation accuracy and stability. The code will be made publicly available at <https://github.com/jingkunchen/scribble-medical-segmentation>.

Index Terms—Scribble, weakly supervised learning, inconsistent labeling, medical image segmentation.

I. INTRODUCTION

MEDICAL image segmentation using deep learning is crucial for medical image analysis. Nonetheless, this task is immensely challenging due to the limited availability of fully annotated data. Segmentation algorithms heavily depend on labeled data for training, but manual labeling is a time-consuming process. Recently, there has been promise in scribble-based segmentation methods, where scribble annotations offer a more efficient means of generating pixel-level annotations. Scribbles require only approximating shape and location information, reducing the labeling workload and enabling non-professionals with basic training to mark the general location of each organ.

Despite the advantages, scribble labeling introduces challenges. Since it is performed manually, personal subjective judgments can lead to labeling bias and inconsistent results. Scribbles within the same class may vary notably, as seen in Fig. 1 with the blue scribbles representing the left ventricle class displaying a progressive decrease in size. Besides size variation, scribbles exhibit diversity in forms, including circular, spiral, and linear shapes. Each shape variation provides unique insights into the organ’s characteristics, emphasizing the diversity within the scribble annotations.

In the case of organs in motion, such as the heart, variations in imaging are significant. Fig. 2 demonstrates differences in cardiac MRI images captured during diastole and systole due to changes in blood flow and tissue characteristics. The complexity of these images complicates scribble labeling, introducing challenges in determining width, intensity, shape, and size. Inconsistencies across different images further hinder the recognition of structures.

Additionally, annotators using scribbles provide only partial information about the object of interest, resulting in essential information loss as scribbles cannot cover all areas of medical images. Such inconsistencies can significantly impact the performance of segmentation models, resulting in suboptimal results for scribble-based approaches.

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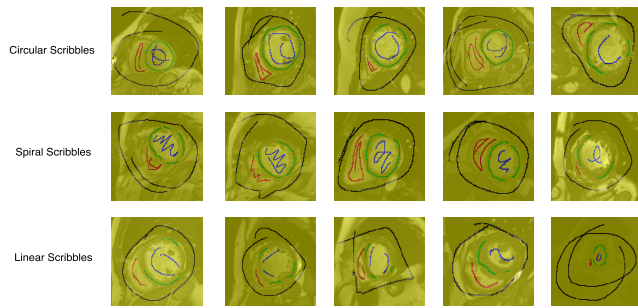


Fig. 1. Blue scribbles for the left ventricle class show a progressive size decrease and feature diverse shapes—circular, spiral, and linear—highlighting both the unique insights and challenges of manual labeling.

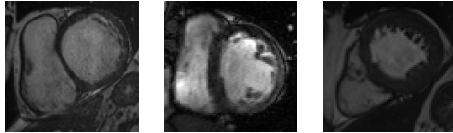


Fig. 2. Heart imaging varies in shape, size, and intensity due to pulsation changes. Differences between diastole and systole complicate scribble labeling, making it challenging to determine width, intensity, shape, and size, and causing inconsistencies across images.

To address these issues, two key questions arise. **Q.1.** How can inconsistent scribble labeling be effectively utilized to improve segmentation network training, considering the influence of personal judgment and morphological changes? **Q.2.** What strategies can be employed to mitigate the problem of essential information loss? Existing scribble-based segmentation techniques aim to tackle these challenges by incorporating additional information such as image context, spatial constraints, and shape priors [1], [2], [3]. However, these methods often suffer from inconsistent labeling outcomes, failing to comprehensively address the problem of essential information loss caused by incomplete and inconsistent scribble annotations.

In this paper, our primary objective is to address the challenges arising from inconsistent scribble annotations and effectively tackle the issue of information loss in medical image segmentation. A comprehensive analysis of various scribble types, as depicted in Fig. 1, indicates that different scribble types can synergistically enhance each other to represent distinct classes more effectively. For instance, the pixel positions marked by various scribbles may not always be identical; some may be located in the center, while others may be positioned along the edges. These diverse annotation forms complement each other. Motivated by this observation, we propose a collaborative approach that integrates varied scribble annotations to comprehensively capture different pixel styles and all relevant information associated with a class.

The integration of scribble annotations provides guidance for images with incomplete and inconsistent scribble annotations. Building on this concept and recognizing the limited pixel coverage in individual image scribble annotations, we create a reference set that captures distinctive pixel characteristics identified by scribble annotations across multiple images, encompassing a wide range of pixel styles for a thorough understanding of the related pixels. The reference set serves as a foundational basis, comprising pixel-level features

used as queries and semantic-level class-tokens obtained from scribbles across multiple images. Leveraging the reference set allows us to align training pixels with their corresponding pixel-level queries and class-tokens in the reference set. This alignment enhances representation smoothness, reducing the impact of noise, variability, and irrelevant information [4]. Additionally, aligning training pixels with semantic-level class-tokens enables the model to capture intrinsic class information, enhancing its ability to recognize different objects, scenes, or concepts in an image [5]. Incorporating both queries and class-tokens from the reference set captures a broader context of features, leading to improved generalization through cross-image guidance.

Concretely, we present a novel method based on pixel-level feature matching to tackle the issues arising from inconsistent scribble labeling and information loss in the context of scribble-based medical image segmentation. Our approach is evaluated on three datasets: ACDC dataset [3], [6], MS-CMRSeg Scribbles dataset [7], and MS-CMRSeg Challenge dataset [8]. Experimental results with five-fold cross-validation on these medical datasets demonstrate the effectiveness of our proposed method, surpassing state-of-the-art methods. The key contributions of this paper can be summarized as follows:

- Create a reference set by incorporating pixel-level queries and class-specific tokens from various cross images. This reference set serves as a benchmark for achieving precise pixel alignment, providing valuable insights for cross-image matching.
- Introduce an improved matching process for scribble-based medical image segmentation, utilizing a reference set to smooth representations and perform class regression for the correction of original features. This refinement leads to an enhancement of the original features, effectively addressing challenges associated with inconsistent labeling and resulting in a significant improvement in segmentation accuracy.
- Alleviate challenges related to annotation bias in training with scribble annotation by establishing a reference set and implementing cross-image matching for pixel alignment. Introducing a pioneering pixel-level feature matching approach in scribble-based medical image segmentation, enhancing model robustness when training with annotations of different style biases. This improves adaptability and reliability in real-world clinical scenarios, contributing significantly to segmentation methodologies.

II. RELATED WORK

Deep learning-based medical image segmentation algorithms have played a significant role in advancing medical practices [9], [10]. However, a major challenge in this field is the limited availability of manual annotations [11], [12], [13]. To address this issue, various strategies, including transfer learning [14], [15], [16], semi-supervised learning [17], [18], [19], [20], and weakly supervised learning [21], [22], [23], have been explored.

Transfer learning aims to improve model performance on smaller datasets by leveraging models pre-trained on larger

datasets. However, it may lead to overfitting on the training set, causing poor generalization on new, unseen data. This challenge is particularly evident in medical imaging due to the complexity of data, such as irregular nodules, lesions, and morphological changes during imaging.

Semi-supervised learning enhances deep learning by incorporating a small amount of labeled data with a larger pool of unlabeled data. Its effectiveness is, however, limited when the assumption of data originating from the same distribution doesn't hold. Moreover, incorporating unlabeled data necessitates detailed annotations by domain experts, incurring substantial costs and requiring constant expert guidance.

In contrast, *weakly supervised learning* provides a flexible solution using scribble annotations, allowing non-experts to label weak annotations after minimal training. This cost-effective approach enhances the efficiency and effectiveness of medical image segmentation, especially when manual annotations are limited or incomplete. In this paper, we focus on weakly supervised learning.

A. Scribble-Based Segmentation

Training segmentation models with scribble annotations offers the potential to significantly reduce annotation workload while maintaining a high level of segmentation accuracy. However, it is crucial to acknowledge that utilizing scribble annotations in training may lead to lower performance compared to using dense annotations. Recent advancements have introduced various methods to address and enhance the performance of segmentation models when incorporating scribble annotations.

The advancement of deep learning has led to the exploration of graphical-based methods, exemplified by the random walk segmentation proposed by [24] and [25]. These methods are designed to emphasize achieving balance within graph models. The main objective is to achieve a more equitable distribution of energy, thereby reducing emphasis on less significant areas during image segmentation. It is noteworthy that while the proposed method was initially developed under full annotation supervision, its applicability extends to scenarios where training involves scribble annotations. To overcome performance challenges associated with scribble annotations, techniques such as minimum entropy regularization [26], specific CRF regularized frameworks [27], and Mumford-Shah regularization [28] have been developed.

In recent years, Liu et al. [29] introduced an uncertainty-aware mean teacher framework that ensures consistent predictions and incorporates a pixel-level uncertainty measure for reliable guidance. Meanwhile, Luo et al. [30] employed a dual-branch network with one encoder and two slightly different decoders. This method dynamically blends their predictions to generate pseudo labels for auxiliary supervision. Zhang et al. [7] proposed CycleMix, integrating mix augmentation with cycle consistency. This approach combines a mixup strategy with random occlusion for supervision augmentation and consistency losses for regularization. ScribbleVC, proposed by Li et al. [31], leverages multimodal information enhancement,

utilizing CNN and Transformer features uniformly, and combines a scribble-based approach with a segmentation network and a class-embedding module. These innovative methods collectively hold significant potential to enhance the performance of scribble-based segmentation approaches and broaden their applicability in domains such as medical imaging and computer vision.

Nevertheless, several challenges have yet to be fully addressed. One prominent challenge is the inconsistency often found in scribble annotations, which may influence the stability and improvement of performance. Additionally, incomplete scribble annotations can lead to information loss [32], hindering the model from accurately and effectively learning all the essential information.

B. Feature Similarity Matching

In recent decades, the increasing demand for accurate analysis of images and videos has spurred notable advancements in methods based on feature analysis. Among these approaches, pixel-level feature matching has emerged as a prominent technique with diverse applications, such as object tracking [33], video segmentation [34], pose estimation [35], and video stabilization [36]. However, most of these techniques calculate how pixels move between frames in a video sequence, focusing on tracing the motion of individual pixels and establishing one-to-one correspondences, commonly known as feature tracking. This approach is different from matching similar pixel-level features between images.

Another type of feature matching technique concentrates on matching overall features at the image level [37]. For instance, content-based image search engines typically analyze an image's entire visual characteristics [38].

However, existing feature-matching methods face several challenges that make them unsuitable for scribble-based medical image segmentation tasks. First, these methods rely heavily on detecting and describing features across the entire image to establish correspondences with other images. This approach assumes the availability of comprehensive visual information, which is often missing in scribble-based annotations. Scribbles provide only sparse and localized information, making it difficult to infer the full range of class characteristics, such as the global texture, shape, and structure of medical regions of interest.

Second, pixel-level matching methods [39], [40] often struggle to handle the unique complexities of medical images. These images frequently exhibit intricate deformations, irregular shapes (e.g., nodules and lesions), and variable textures due to both pathological variability and imaging artifacts. Such variations challenge the robustness of feature matching, especially when relying on limited scribble annotations that do not adequately represent these irregularities.

Lastly, the performance of pixel-level feature matching methods is highly dependent on the quality and quantity of the scribbles. High-quality annotations with sufficient coverage are necessary to extract meaningful features, but in practice, scribbles are often sparse and incomplete, leading to suboptimal feature representation. Existing methods frequently overlook

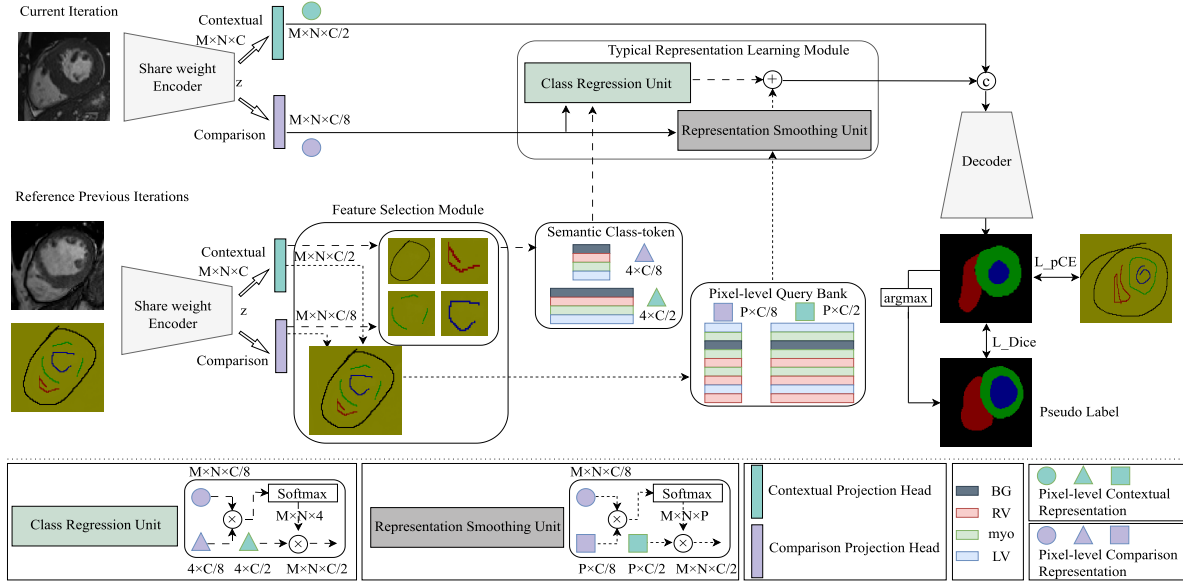


Fig. 3. The framework consists of key components: an encoder for feature extraction from input images, a decoder for generating a segmentation map, and a feature selection module to create a reference set with pixel-level representations and semantic class-tokens. A representation learning module connects the encoder and decoder, performing regression and smoothing operations by matching training pixels with semantic class-tokens and representations. The final loss function includes pCE loss based on scribble annotations and DICE loss based on pseudo-labels.

this dependency, failing to incorporate strategies to compensate for the inherent limitations of scribble annotations.

III. METHOD

The proposed framework (Fig. 3) consists of a single encoder and decoder connected through a representation learning module. The encoder extracts features, which are then sent to two MLP heads for dimensionality reduction: a contextual projection MLP head (in green) reduces the feature channel by half, preserving image information, termed as contextual features. Simultaneously, a comparison projection MLP head (in purple) reduces the feature channel to one-eighth to minimize computation costs for efficient comparisons, known as comparison features.

To create a comprehensive reference set with diverse pixel-level features, we introduce a feature selection module that extracts feature points from the representations obtained via the contextual MLP head and the comparison MLP head.

As is shown in Fig. 4, these feature points are identified using both *class-specific scribbles* (with specific class information to calculate class-tokens) and *all scribbles* (to capture all scribble-located features as queries). This allows extraction of relevant feature points and creation of class-specific and all scribble-located representations, forming a reliable reference set (with class-tokens and pixel-level queries) across different images.

For class-tokens, our approach involves formulating global distinct prototypes for each class, enabling the integration of class information into class-specific representations. These tokens are pivotal in guiding the training of unlabeled pixels by providing a tailored prototype for each class, establishing a robust guiding mechanism throughout the training process.

In addressing potential loss of local information due to a global prototype, we utilize all pixels annotated with scribbles

as local queries for the queries component. These local queries serve as a foundational reference for training pixels, aiming to alleviate the potential loss of local information associated with global class-tokens. The simultaneous utilization of both class-tokens and pixel-level queries contributes to a nuanced and logically structured pixel-matching training paradigm, enhancing the sophistication of the learning process. *Importantly, it's noteworthy that both class-tokens and pixel-level queries possess their own contextual features and comparison features.*

A novel module, the typical representation learning module, leverages class-tokens and pixel-level representations in queries from the reference set for cross-image guidance. This module includes a representation smoothing unit and a class regression unit. The former employs feature smoothing with queries to revise training pixels into a more informative representation, reducing noise and irrelevant information. The latter uses class-tokens for regression, providing a compact and expressive representation of semantic content. Smoothing and regression outputs are combined with contextual representations from training images and fed into the decoder for segmentation.

In terms of loss design, we incorporate the pixel-wise Cross-Entropy (pCE) loss based on scribble annotations and the Dice (DICE) loss based on pseudo labels [30], [41], respectively. Notably, our approach maintains the original encoder and decoder architecture without requiring collaboration between multiple networks, ensuring superior efficiency and yielding more accurate segmentation outcomes.

A. Reference Set

To establish a diverse reference set of pixel styles and class-specific prototype for reference, we designed a feature selection module, as depicted in Fig. 4. This module leverages

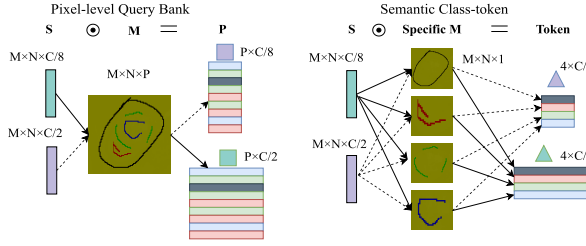


Fig. 4. The feature selection module uses scribbles (Pixel level Queries) are extracted by scribble maps M and semantic class-tokens are extracted by the class-specific scribble maps **specificM** to locate pixels. It extracts features from these positions to create a reference set, which includes semantic class-tokens and pixel-level queries. Both representations have comparison and contextual elements.

scribbles to extract informative features from images. Class-specific and general scribble maps, scaled to the size of the corresponding feature map, are combined with contextual and comparison feature maps to extract features at scribble positions, leading to the generation of pixel-level contextual and comparison representations. The left side of Fig. 4 illustrates the extraction of features at all scribble positions, stored in queries to enrich the variety of pixel-level representations. These queries serve as guidance for the subsequent smoothing process. On the right side of Fig. 4, the use of class-specific scribble maps is demonstrated for extracting class-specific representations. These representations are then utilized to calculate prototypes (class-tokens) for each class, playing a pivotal role in training pixel regression.

1) *Pixel-Level Queries*: The process for extracting scribble-based features is described in Eq. 1 and Fig. 4, where P ($P \times C/8$ and $P \times C/2$) represents matrices with extracted pixels, S ($M \times N \times P$) represents the rescaled scribble map adjusted to the size of the corresponding feature map, M ($M \times N \times C/8$ and $M \times N \times C/2$) represents the contextual and comparison features from the two MLP heads, and $\odot$ is the Hadamard product of S and M . We utilize the Hadamard product to extract pixel values at scribble locations by multiplying a rescaled scribble map with a feature map of the same size. The specific process is as follows:

$$P = S \odot M, \quad (1)$$

We use a specialized queue called the pixel-level queries to efficiently store and update queries. It records a predetermined set number of localized pixel-level features derived from input scribbles. The queue operates on the First In First Out (FIFO) principle, processing and removing pixels in the order they were added from different training iterations. This implies that features placed in the queue during the previous batch training process will be replaced by the latest features in the next update, creating a streamlined replacement mechanism. The FIFO mechanism updates representations with the latest pixel-level features, ensuring query relevance and accuracy during the model's learning.

2) *Semantic-Level Class-Tokens*: The extraction of class-specific semantic tokens is predicated on sparse pixel features derived from class-specific scribbles, which encapsulate various classes representations. This extraction process parallels pixel-level queries but incorporates distinct

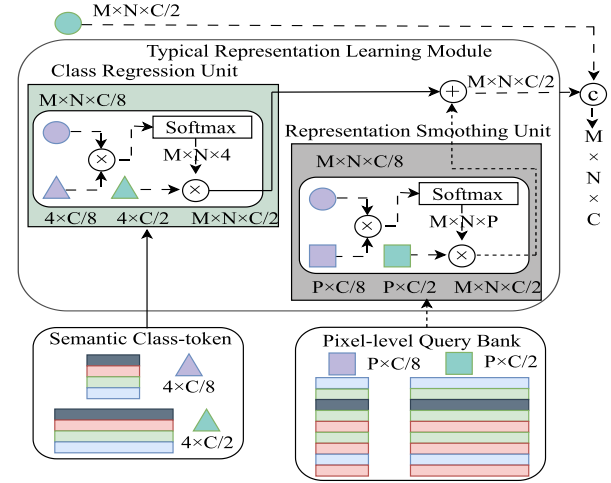


Fig. 5. The typical representation learning module includes a class regression unit and a representation smoothing unit, both using semantic class-tokens and pixel-level queries for regression and smoothing operations. The process involves using the Class Regression Unit and Representation Smoothing Unit for matching. It compares training pixels (purple circles) with a reference set (purple triangles and squares) using matrix multiplication. The Softmax function assesses similarity, and then contextual representations from the reference set (green triangle and square) are multiplied with the similarity results to refine the training pixels using the reference set.

methodologies for storage and updates. Specifically, class-specific scribble maps are employed to extract pixel-level features unique to each class. These features are subsequently averaged to compute the mean value for each class, culminating in the formation of semantic-level class tokens. Throughout the training process, each class is represented by a single class token. The update of the class tokens m is facilitated by utilizing the mean value θ of all pixel values at the scribble positions for each class in every iteration. This updating mechanism is elucidated in Eq. 2.

$$m_{t+1} = f(\theta_t, m_t). \quad (2)$$

In the experiment, we utilize two moments: m_t , the class token for the current iteration, and θ_t , the mean value for the same iteration. We combine these using a function f to compute the moment update at time $t + 1$. The function f can be configured in various ways. In our approach, we define f as follows: the mean value at time $t + 1$ is obtained by multiplying m_t by 0.99 and adding θ_t multiplied by 0.01. This strategy enhances the resilience and dependability of class-tokens, particularly when dealing with inconsistent and incomplete scribble inputs.

By using this moment update, we handle data variations for a stable and reliable learning process. It is particularly effective with incomplete or inconsistent scribbles, ensuring a consistent and accurate learning experience for the model.

B. Typical Representation Learning Module

Fig. 5 illustrates a representation learning module with two key components: the representation smoothing unit and the class regression unit. This module aligns pixel-level queries and semantic class-tokens from the reference set through matching operations based on comparison (short rectangles) and contextual (long rectangles) representations within

class-tokens and queries. The matching processes within the class regression unit and representation smoothing unit work together to facilitate effective feature smoothing and regression. The representation smoothing unit plays a crucial role in aligning the pixel-level features of training pixels with pixel-level queries, ensuring a refined enhancement of the pixel-level features. Simultaneously, the class regression unit uses the aligned representations to capture class-specific information, enabling accurate regression for each class.

$$X_S = \text{softmax}(X_C \otimes Q_C^T) \otimes Q_{CTX}. \quad (3)$$

Eq. 3 outlines the process of the representation smoothing unit. In this equation, X_C (depicted as a purple circle) represents the pixel-level feature generated by the comparison MLP layer of the training pixels. Q_C (represented as a purple square) and Q_{CTX} (shown as a green square) represent the comparison representations and contextual representations, respectively, stored in the queries. The $\otimes$ operation signifies matrix multiplication, capturing semantic relationships between the features in X_C and Q_C . Specifically, matrix multiplication is performed between X_C and Q_C , resulting in a revised feature map that highlights regions of X_C exhibiting similarity to Q_C .

The Softmax function is applied along the channel dimension of the feature map to refine features based on contextual information. This generates a probability distribution over the channels, indicating the relative importance of each channel in relation to the training pixels. The resulting Softmax weights act as a weight map, assigning higher weights to more relevant channels. Finally, a matrix multiplication blends the weighted feature map with contextual representations Q_{CTX} , enhancing contextual relevance for the training pixels.

However, relying solely on pixel-level features may overlook global semantic guidance. To address this issue, we incorporate semantic-level class-tokens, which provide class-specific semantic guidance. Similar to pixel-level queries, these tokens assist in the matching process with the training pixels. After matching, we compute the similarity between the training pixels and each class-specific token. This similarity is used in the regression process to produce regressed pixel-level features (X_R) that are contextually pertinent to a specific class-tokens. The regression process is represented by Eq. 4, where X_C represents the comparison representations of the training pixels, T_C is the comparison class-tokens, and T_{CTX} is the contextual class-tokens. In Fig. 5, the purple triangle (T_C) represents comparison class-tokens, while the green triangle (T_{CTX}) represents contextual class-tokens. The regressed pixel-level features that are contextually relevant to a given class-tokens are denoted as X_R .

$$X_R = \text{softmax}(X_C \otimes T_C^T) \otimes T_{CTX}, \quad (4)$$

The final revised feature maps involve the integration of two distinct representations from the two units. This integration is achieved through an addition operation, combining the two feature maps to produce a cross-image scribble-guided revised feature, as shown in Fig. 5.

Subsequently, the combined features are concatenated with the contextual features from the training pixels, as illustrated

on the right side of Fig. 5 and expressed by Eq. 5, where X_S represents the smoothed pixel-level features, X_{QCT} denotes the contextual representations of the training pixels, and X_E represents the revised feature. By adding X_R and X_S together and concatenating them with X_{QCT} , we obtain the final feature X_E , allowing these features to be guided and refined by pixels supervised with scribbles from different images. This revised feature is then fed into the decoder to generate the segmentation map. The purpose of this approach is to retain half of the channels preserving the original encoder output features, while the other half is corrected through guidance from pixels supervised with scribbles from different images (class-tokens and queries).

$$X_E = \text{concatenate}(X_R + X_S, X_{QCT}), \quad (5)$$

C. pCE Loss and Pseudo-Labeling Based DICE Loss

In our segmentation network, we use two types of supervision: scribble-based supervision and pseudo-labeling supervision. For scribble-based supervision, we employ the partial Cross-Entropy (CE) loss, denoted as $\mathcal{L}_{pCE}$. This loss is designed for handling scribble annotation and is defined in Eq. 6. Here, N is the number of pixels in the scribbles, C is the number of classes, y is the label tensor, and $y_{i,c}$ and $\hat{y}_{i,c}$ represent pixel-to-class associations in the scribble and predicted map, respectively.

$$\mathcal{L}_{pCE}(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}), \quad (6)$$

We use scribble-based supervision and incorporate the pseudo-labeling technique $\mathcal{L}_{Dice}$ [30]. It calculates the Dice loss (Eq. 7) based on predicted probabilities $\hat{y}_i$ for each pixel (i) in the target class. The weight (w_i) is assigned using the argmax function on the probability map, giving the highest probability class a weight of 1 and others 0. To avoid division by zero, we add a small constant value ϵ to the denominator when the predicted and ground truth masks are empty.

$$\mathcal{L}_{Dice}(\hat{y}) = 1 - \frac{2 \sum_{i=1}^N w_i \hat{y}_i}{\sum_{i=1}^N w_i \hat{y}_i + \sum_{i=1}^N \hat{y}_i + \epsilon}, \quad (7)$$

To train our network with both types of supervision, we combine the losses using hyperparameters α with a fixed weight ratio of 0.5 (based on [30]). The total loss, denoted as $\mathcal{L}$, is expressed in Eq. 8, where α control the relative importance of the two losses, $\mathcal{L}_{pCE}$ and $\mathcal{L}_{Dice}$ (partial CE loss and DICE loss, respectively).

$$\mathcal{L} = \mathcal{L}_{pCE} + \alpha \times \mathcal{L}_{Dice}, \quad (8)$$

IV. EXPERIMENT

A. Experimental Setup

The intricacy of cardiac images and the variability in scribbles pose challenges in the segmentation task. This unique nature emphasizes the difficulty of training models using scribble annotations, prompting our choice of cardiac databases. To systematically evaluate our method, we conducted experiments on three diverse cardiac datasets: ACDC,



Fig. 6. ACDC Scribble dataset (left), MS-CMRSeg Scribbles dataset (middle), and MS-CMRSeg Challenge dataset with algorithm-generated scribble annotations (right).

MS-CMRSeg Scribbles, and MS-CMRSeg Challenge. Fig. 6 provides visual representations of sample images from these datasets, illustrating the diversity and complexity involved in our experimental evaluation.

The ACDC and MS-CMRSeg Scribbles datasets are commonly used in scribble-based medical image segmentation tasks, offering valuable annotations. For the MS-CMRSeg Challenge dataset, we employed the annotation generation algorithm proposed by [30], resulting in annotations with fewer pixels compared to manual annotations.

The ACDC dataset, recognized as a benchmark for cardiac image segmentation [3], comprises two modalities of cardiac MRI images from 100 patients (200 3D scans) captured by 1.5T and 3T scanners. The segmentation task involves identifying the left ventricle (LV), myocardium (myo), and right ventricle (RV) in approximately 256×256 -pixel resolution images. Scribble annotations are provided by [6], with mutually exclusive training and testing sets based on patient IDs.

The MS-CMRSeg Scribbles dataset consists of 25 3D late gadolinium enhancement (LGE) MS-CMR images with scribble annotations at a higher resolution of 480×480 compared to the ACDC dataset. Each image in this dataset is annotated with four classes (Background, RV, myo, and LV).

Lastly, the MS-CMRSeg Challenge dataset, valuable for cardiac image segmentation research, initially includes 135 3D images from three modalities (LGE, bSSFP, and T2) from 45 patients, each with a resolution of 480×480 pixels. Due to poor image quality and limited slices in the T2 modality, only 90 3D images from the LGE and bSSFP modalities are used. The dataset involves three classes: RV, myo, and LV. The annotations generated using the method from [30] for this dataset contain fewer annotated pixels compared to ACDC and MS-CMRSeg Scribbles datasets (Fig. 6).

For hyperparameter selection, we adhered to the same configuration for the loss weight as outlined in [30], where α was set to 0.5. The number of class-tokens corresponds to the number of classes, and each class in the queries is equipped to store 1000 features.

Our method was benchmarked against a range of approaches, including pCE [41], RW [24], USTM [29], S2L [42], MLoss [28], EM [26], RLoss [27], MPLS [30], and ScribbleVC [31]. Furthermore, deliberately inducing a higher degree of bias in scribble annotations, we conducted a comparative analysis with the MPLS method [30]—recognized as the second-best performer on the ACDC and MS-CMRSeg Scribbles datasets. This comparison aimed to evaluate the

TABLE I
EVALUATION ON THE ACDC SCRIBBLE DATASET. BOLD IS BEST, FOLLOWED BY UNDERLINE AND WAVY UNDERLINE

Method	DSC $\uparrow$			
	RV	myo	LV	mean
FullSup [43]	0.882 $\pm$ 0.095	0.883 $\pm$ 0.042	0.930 $\pm$ 0.074	0.898 $\pm$ 0.070
pCE [41]	0.625 $\pm$ 0.160	0.668 $\pm$ 0.095	0.766 $\pm$ 0.156	0.686 $\pm$ 0.137
RW [24]	0.813 $\pm$ 0.113	0.708 $\pm$ 0.066	0.844 $\pm$ 0.091	0.788 $\pm$ 0.090
USTM [29]	0.815 $\pm$ 0.115	0.756 $\pm$ 0.081	0.785 $\pm$ 0.162	0.786 $\pm$ 0.119
S2L [42]	0.833 $\pm$ 0.103	0.806 $\pm$ 0.069	0.856 $\pm$ 0.121	0.832 $\pm$ 0.098
MLoss [28]	0.809 $\pm$ 0.093	0.832 $\pm$ 0.055	0.876 $\pm$ 0.093	0.839 $\pm$ 0.080
EM [26]	0.839 $\pm$ 0.108	0.812 $\pm$ 0.062	0.887 $\pm$ 0.099	0.846 $\pm$ 0.089
RLoss [27]	0.856 $\pm$ 0.101	0.817 $\pm$ 0.054	0.896 $\pm$ 0.086	0.856 $\pm$ 0.080
MPLS [30]	0.861$\pm$0.096	0.842 $\pm$ 0.054	0.913 $\pm$ 0.082	0.872 $\pm$ 0.077
ScribbleVC [31]	0.853 $\pm$ 0.087	0.839 $\pm$ 0.054	0.902 $\pm$ 0.092	0.865 $\pm$ 0.061
ours	0.860 $\pm$ 0.086	0.855$\pm$0.047	0.923$\pm$0.069	0.879$\pm$0.051
Method	95HD $\downarrow$			
	RV	myo	LV	mean
FullSup [43]	6.9 $\pm$ 10.8	5.9 $\pm$ 15.2	8.1 $\pm$ 20.9	7.0 $\pm$ 15.6
pCE [41]	187.2 $\pm$ 35.2	165.1 $\pm$ 34.4	167.7 $\pm$ 55.0	173.3 $\pm$ 41.5
RW [24]	11.1 $\pm$ 17.3	9.8 $\pm$ 8.9	9.2 $\pm$ 13.0	10.0 $\pm$ 13.1
USTM [29]	54.7 $\pm$ 65.7	112.2 $\pm$ 54.1	139.6 $\pm$ 57.7	102.2 $\pm$ 59.2
S2L [42]	14.6 $\pm$ 30.9	37.1 $\pm$ 49.4	65.2 $\pm$ 65.1	38.9 $\pm$ 48.5
MLoss [28]	17.1 $\pm$ 30.8	28.2 $\pm$ 43.2	37.9 $\pm$ 59.6	27.7 $\pm$ 44.5
EM [26]	25.7 $\pm$ 44.5	47.4 $\pm$ 50.6	43.8 $\pm$ 57.6	39.0 $\pm$ 50.9
RLoss [27]	7.9 $\pm$ 12.6	6.0 $\pm$ 6.9	7.0 $\pm$ 13.5	6.9 $\pm$ 11.0
MPLS [30]	7.9 $\pm$ 12.5	9.7 $\pm$ 23.2	12.1 $\pm$ 27.2	9.9 $\pm$ 21.0
ScribbleVC [31]	6.7$\pm$8.8	4.3$\pm$4.0	5.2$\pm$8.5	5.4$\pm$5.2
ours	8.9 $\pm$ 12.6	6.6 $\pm$ 18.0	6.6 $\pm$ 17.3	7.3 $\pm$ 12.4
P-values	< 5e-2 (DSC), < 5e-2 (95HD)			

robustness and advancements of our approach under heightened bias conditions, as elaborated in Section V-A.

For a comprehensive evaluation, we employed an 80% training and 20% testing split for five-fold cross-validation on each dataset. Two widely-used metrics, Dice Similarity Coefficient (DSC) and 95th percentile of Hausdorff Distance (95HD), were utilized. DSC measures the overlap between predicted and ground truth segmentations, while 95HD measures the maximum distance between the predicted and ground truth. To assess the statistical significance of observed variations between methods, we conducted Wilcoxon rank-sum tests to compute corresponding p-values.

B. Comparison to the State-of-the-Art

1) *ACDC Dataset*: Tab. I provides an overview of the evaluation results for various methods on the ACDC Scribble Dataset, presenting performance metrics in terms of DSC and 95HD across different cardiac structures. The table facilitates a comparison of our proposed method with state-of-the-art techniques, utilizing bold, underline, and wavy underline to denote the best, second-best, and third-best performances, respectively.

Our method outperforms MPLS, the second-best approach, with a significant 1.3% improvement in the myocardium (myo) and a 1% enhancement in the left ventricle (LV) in terms of DSC. Despite ranking second for the right ventricle (RV) class, our method achieves the highest average DSC across all classes, surpassing the second-best method by 0.7%. These results underscore the efficacy of our approach in achieving superior segmentation accuracy for cardiac structures.

For the 95HD metric, our method demonstrates competitive performance, with results 1.9 higher than the best-performing method (ScribbleVC) and 0.4 higher than the second-best method (RLoss), ranking third overall.

However, our observations suggest that the 95HD metric may not be well-suited for models trained with weak (scribble)

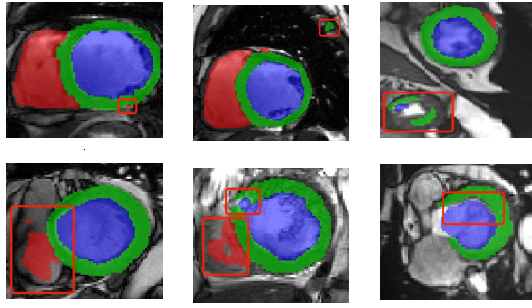


Fig. 7. The first row emphasizes a notable rise in 95HD metric values within models trained using scribble annotations. Minor errors in these models exert a significant impact on the overall evaluation, jeopardizing the assessment of performance and indicating potential inadequacies for situations involving weak (scribble) annotation training. Conversely, the second row demonstrates that, although 95HD results may not show a significant decrease, the actual segmentation outcomes could be suboptimal. In these instances, the DICE coefficient, which utilizes overlap as a criterion, consistently offers a more reliable performance assessment.

annotations, as evidenced by the significantly elevated 95HD values in scribble-based models shown in the first row of Fig. 7. These subtle anomalies underscore the disproportionate impact of minor errors on the overall evaluation, potentially influencing the performance assessment negatively.

In contrast, the second row of Fig. 7 demonstrates that, while 95HD results may not show a significant deterioration, the actual segmentation outcomes may be suboptimal. In such instances, the DICE coefficient, utilizing overlap as a measuring criterion, consistently provides a relatively reliable performance assessment.

The p-values in Tab. I indicate statistical significance, all being less than $5e-2$. This suggests that the observed differences in performance between our method and the compared methods are statistically significant. These low p-values reinforce the assertion that our method's performance is not due to random chance but represent genuine improvements over the compared methods. The statistical significance further strengthens the credibility of our method's efficacy in the multi-class segmentation of cardiac structures.

2) *MS-CMRseg Scribbles Dataset*: In the evaluation of the MS-CMRseg Scribbles dataset, our proposed method demonstrates superior performance compared to other state-of-the-art techniques. The mean Dice Similarity Coefficient (DSC) achieved by our method is 85.3%, outperforming the closest competitor by 1.5%. The competitiveness of our approach extends across different cardiac structures, including the right ventricle (RV), myocardium (myo), and left ventricle (LV), as indicated by the respective DSC values. Notably, our method excels in RV segmentation with a DSC of 84.5% and myo segmentation with a DSC of 80.9%, surpassing other methods. Our method ranks as the second-best performer for left ventricle (LV) segmentation, achieving a notable DSC of 90.5%.

The observed elevation in the 95th percentile of the Hausdorff Distance (95HD) within our findings (with an average value of 41.4, ranked fourth) intimates intermittent substantial spatial discordance between our segmentation outcomes and the corresponding ground truth. This phenomenon may be attributed, in part, to the inherent limitations of the supervisory

TABLE II
EVALUATION ON THE MS-CMRSEG SCRIBBLES DATASET

Method	DSC $\uparrow$			
	RV	myo	LV	mean
FullSup [43]	0.858 $\pm$ 0.108	0.837 $\pm$ 0.058	0.917 $\pm$ 0.049	0.871 $\pm$ 0.06
pCE [41]	0.599 $\pm$ 0.151	0.626 $\pm$ 0.083	0.790 $\pm$ 0.08	0.672 $\pm$ 0.075
RW [24]	0.825 $\pm$ 0.141	0.795 $\pm$ 0.073	0.891 $\pm$ 0.059	0.837 $\pm$ 0.075
USTM [29]	0.653 $\pm$ 0.166	0.742 $\pm$ 0.070	0.823 $\pm$ 0.067	0.740 $\pm$ 0.085
S2L [42]	0.811 $\pm$ 0.117	0.806 $\pm$ 0.064	0.887 $\pm$ 0.060	0.835 $\pm$ 0.066
MLoss [28]	0.776 $\pm$ 0.170	0.758 $\pm$ 0.083	0.855 $\pm$ 0.071	0.797 $\pm$ 0.097
EM [26]	0.801 $\pm$ 0.148	0.796 $\pm$ 0.064	0.885 $\pm$ 0.055	0.827 $\pm$ 0.074
RLoss [27]	0.807 $\pm$ 0.177	0.800 $\pm$ 0.075	0.903 $\pm$ 0.053	0.836 $\pm$ 0.095
MPLS [30]	0.813 $\pm$ 0.183	0.791 $\pm$ 0.083	0.910$\pm$0.052	0.838 $\pm$ 0.090
ScribbleVC [31]	0.808 $\pm$ 0.181	0.770 $\pm$ 0.090	0.905 $\pm$ 0.048	0.827 $\pm$ 0.095
ours	0.845$\pm$0.104	0.809$\pm$0.060	0.905 $\pm$ 0.046	0.853$\pm$0.062
Method	95HD $\downarrow$			
	RV	myo	LV	mean
FullSup [43]	8.2 $\pm$ 12.5	2.5 $\pm$ 1.2	9.2 $\pm$ 18.3	6.6 $\pm$ 8.5
pCE [41]	221.3 $\pm$ 28.8	176.1 $\pm$ 36.3	160.1 $\pm$ 46.1	185.8 $\pm$ 27.2
RW [24]	13.4 $\pm$ 34.8	7.7 $\pm$ 13.3	16.4 $\pm$ 33.8	12.5 $\pm$ 23.4
USTM [29]	171.3 $\pm$ 76.5	130.8 $\pm$ 64.4	151.9 $\pm$ 49.6	151.3 $\pm$ 55.0
S2L [42]	113.4 $\pm$ 78.0	47.1 $\pm$ 62.4	90.4 $\pm$ 7.7	83.6 $\pm$ 44.1
MLoss [28]	66.8 $\pm$ 77.8	121.8 $\pm$ 72.0	85.9 $\pm$ 67.8	91.5 $\pm$ 52.9
EM [26]	93.9 $\pm$ 73.4	68.1 $\pm$ 62.6	97.4 $\pm$ 7.0	86.5 $\pm$ 46.3
RLoss [27]	8.3$\pm$12.8	6.4 $\pm$ 11.6	11.3 $\pm$ 18.2	8.7 $\pm$ 12.2
MPLS [30]	50.6 $\pm$ 73.6	48.22 $\pm$ 76.5	27.7 $\pm$ 44.8	42.2 $\pm$ 47.1
ScribbleVC [31]	10.8 $\pm$ 25.2	3.4$\pm$2.2	6.8$\pm$14.7	7.0$\pm$10.1
ours	33.8 $\pm$ 56.8	42.1 $\pm$ 57.6	48.2 $\pm$ 60.9	41.4 $\pm$ 43.7
P-values	< 5e-2 (DSC), < 5e-2 (95HD)			

information derived from scribble annotation. The intrinsically weak delineation provided by scribbles, particularly in the nuanced characterization of boundaries, constitutes a contributing factor. Such limitations pose challenges in the accurate demarcation of intricate cardiac structures, exacerbated by the inherent noise prevalent in certain regions of the medical imagery. Notwithstanding these challenges, our approach maintains a competitive edge, primarily attributed to its consistently superior performance in terms of the Dice Similarity Coefficient (DSC), which attests to its robust overall segmentation accuracy. The algorithm's proficiency in handling diverse cardiac structures, coupled with its resilience across varying image resolutions, underscores its reliability.

The robustness of our method is further highlighted by its consistent performance across various image resolutions. This is evident from the results presented in Tab. II, where our approach consistently achieves superior outcomes. The statistical significance of our results is supported by p-values below $5e-2$ for both DSC and 95HD metrics, confirming the reliability of the observed performance improvements.

In summary, our proposed method surpasses existing approaches in the segmentation of cardiac structures, particularly excelling in LV segmentation. The consistent and superior performance across different evaluation metrics and image resolutions establishes the effectiveness and reliability of our method for the MS-CMRseg Scribbles dataset.

3) *MS-CMRseg Challenge Dataset*: The results presented in Tab. III showcase the superior performance of our proposed method on the MS-CMRseg Challenge Dataset, where scribble annotations contain fewer pixels. Our method demonstrates a significant advancement with a notable mean DICE improvement of 2.4% over the best-performing previous method.

Specifically, our method achieves a DSC of 72.7% for the right ventricle (RV), 65.2% for the myocardium (myo), and an impressive 82.6% for the left ventricle (LV), resulting in a mean DSC of 73.5%. These results highlight the potential of our method in addressing the inherent challenge of essential

TABLE III
EVALUATION ON MS-CMRSEG CHALLENGE DATASET

Method	DSC↑			
	RV	myo	LV	mean
FullSup [43]	0.883±0.058	0.856±0.042	0.936±0.034	0.892±0.035
pCE [41]	0.694±0.076	0.620±0.077	0.797±0.078	0.704±0.049
RW [24]	0.754±0.081	0.503±0.060	0.342±0.142	0.533±0.065
USTM [29]	0.700±0.089	0.617±0.077	0.773±0.068	0.697±0.051
S2L [42]	0.678±0.090	0.597±0.071	0.749±0.077	0.675±0.047
MLoss [28]	0.646±0.085	0.724±0.073	0.417±0.103	0.596±0.053
EM [26]	0.723±0.085	0.633±0.076	0.776±0.070	0.711±0.050
RLoss [27]	0.810±0.100	0.631±0.079	0.529±0.251	0.657±0.112
MPLS [30]	0.719±0.105	0.622±0.083	0.754±0.100	0.699±0.074
ScribbleVC [31]	0.828±0.105	0.660±0.067	0.636±0.138	0.708±0.075
ours	<u>0.727±0.083</u>	<u>0.652±0.078</u>	0.826±0.059	0.735±0.046
Method	95HD↓			
	RV	myo	LV	mean
FullSup [43]	6.7±14.6	4.1±8.2	5.4±12.2	5.4±7.7
pCE [41]	49.3±26.8	29.6±18.6	11.3±15.9	30.1±13.1
RW [24]	9.9±10.9	11.5±6.5	19.4±10.2	13.6±5.9
USTM [29]	34.3±20.7	31.9±18.2	15.2±18.9	27.1±12.8
S2L [42]	36.2±26.7	30.1±16.6	16.1±18.0	27.5±14.1
MLoss [28]	35.7±27.6	28.2±19.3	20.9±14.1	28.2±14.2
EM [26]	28.9±20.9	26.4±18.2	12.0±15.4	22.4±11.5
RLoss [27]	12.2±18.7	12.3±12.4	13.9±10.8	12.8±9.3
MPLS [30]	34.2±26.0	26.9±18.7	22.5±21.2	27.9±18.0
ScribbleVC [31]	6.5±11.8	6.5±4.2	9.6±4.4	7.5±4.8
ours	33.8±23.0	15.5±15.7	8.3±12.6	19.2±11.9
P-values	< 5e-2 (DSC), < 5e-2 (95HD)			

information loss in scribbles, showcasing its ability to accurately segment multiple cardiac structures using only a small number of pixels.

Regarding the 95HD, while our method may not exhibit the best performance in every category, it remains competitive. Specifically, our approach achieves 33.8 for RV, 15.5 for myo, and 8.3 for LV, resulting in a mean 95HD of 19.2. Although it may not be the best in every aspect, our method remains a strong contender.

The statistical significance of our method's superiority is supported by p-values less than 5e-2 for both DSC and 95HD, affirming its robustness and effectiveness. These findings emphasize the potential of our method to provide valuable guidance for medical practitioners by accurately segmenting cardiac structures, even with the limited information inherent in scribble annotations with fewer pixels.

V. ABLATION STUDIES

A. The Influence of Increasing Scribble Bias on Performance Enhancement

To assess the effectiveness of our method under increasing scribble bias, experiments were conducted using the ACDC dataset. Bias was systematically introduced by replacing portions of manual scribbles with algorithmically generated ones from [30]. Specifically, 30% and 50% of manual scribbles were replaced, designating cases 1-70 (1-50) for manual scribbles and cases 71-100 (51-100) for generated ones. The evaluation encompassed three settings: 100% manual annotations, 70% manual combined with 30% generated annotations, and 50% manual combined with 50% generated annotations. The comparison employed five-fold cross-validation and compared with the second-best methods in the ACDC Scribble dataset and MS-CMRseg Scribbles Dataset.

Tab. IV presents comparative results, illustrating the performance enhancement achieved by each method under increasing scribble bias. Our method consistently outperforms

TABLE IV
EVALUATION WITH A COMBINATION OF MANUALLY AND GENERATED SCRIBBLES (50%/50%, 70%/30%, 100%/0%)

50/50	DSC↑			
	RV	myo	LV	mean
MPLS	0.819±0.107	0.806±0.064	0.893±0.088	0.839±0.062
Ours	0.836±0.100	0.821±0.061	0.902±0.083	0.853±0.057
70/30	DSC↑			
	RV	myo	LV	mean
MPLS	0.837±0.101	0.817±0.067	0.900±0.075	0.851±0.059
Ours	0.844±0.091	0.833±0.058	0.909±0.083	0.862±0.057
100/0	DSC↑			
	RV	myo	LV	mean
MPLS	0.861±0.096	0.842±0.054	0.913±0.082	0.872±0.077
Ours	<u>0.860±0.086</u>	0.855±0.047	0.923±0.069	0.879±0.051

TABLE V
EFFECTIVENESS EVALUATION OF PIXEL-LEVEL QUERIES (PQ)

Method	DSC↑			
	RV	myo	LV	mean
ACDC				
FullSup	0.882±0.095	0.883±0.042	0.930±0.074	0.898±0.070
pCE	0.625±0.160	0.668±0.095	0.766±0.156	0.686±0.137
W/O PQ	0.852±0.095	0.847±0.053	0.913±0.086	0.871±0.061
Ours	0.860±0.086	0.855±0.047	0.923±0.069	0.879±0.051
MS-CMRSeg				
FullSup	0.883±0.058	0.856±0.042	0.936±0.034	0.892±0.035
pCE	0.694±0.076	0.620±0.077	0.797±0.078	0.704±0.049
W/O PQ	0.716±0.088	0.628±0.075	0.804±0.065	0.716±0.048
Ours	0.727±0.083	0.652±0.078	0.826±0.059	0.735±0.046

MPLS in all three classes, exhibiting mean DSC improvements of 0.7%, 1.6%, and 0.9% with 70% manual annotations and 30% generated annotations. In the setting of 50% manual annotations and 50% generated annotations, our method demonstrates superiority over MPLS, with improvements of 1.7%, 1.5%, and 0.9% in the three classes.

As annotation bias intensifies (from 0% to 30% and then 50%), both MPLS and our method experience a performance decline. MPLS decreases from 87.2% to 83.9%, and our method decreases from 87.9% to 85.3%. This decline is attributed to additional challenges introduced by increasing annotation bias. However, our method consistently outperforms MPLS by 0.7% to 1.4%, and with the increase in bias, its performance decline is minimal. This indicates that *as bias increases, our proposed method exhibits better robustness compared to other methods.*

This study validates the effectiveness of our method, especially in situations with increased annotation bias. It consistently outperforms state-of-the-art approaches, showcasing its potential to improve performance even under highly biased annotations.

B. The Impact of Pixel-Level Queries

We evaluated the impact of pixel-level queries using five-fold cross-validation, comparing the DSC with and without their usage. Results in Tab. V demonstrate a notable improvement in segmentation performance when utilizing the queries. Specifically, on the ACDC dataset, DSC increased from 87.1% to 87.9%. On the MS-CMRSeg Challenge dataset, DSC increased from 71.6% to 73.5%. The results show substantial enhancements for each class in both databases. The integration of pixel-level queries in scribble-based segmentation tasks significantly enhances segmentation accuracy.

TABLE VI

EFFECTIVENESS EVALUATION OF SEMANTIC-LEVEL CLASS-TOKENS (CT)

Method	DSC $\uparrow$			
	RV	myo	LV	mean
ACDC				
FullSup	0.882 $\pm$ 0.095	0.883 $\pm$ 0.042	0.930 $\pm$ 0.074	0.898 $\pm$ 0.070
pCE	0.625 $\pm$ 0.160	0.668 $\pm$ 0.095	0.766 $\pm$ 0.156	0.686 $\pm$ 0.137
W/O CT	0.855 $\pm$ 0.092	0.847 $\pm$ 0.051	0.914 $\pm$ 0.079	0.872 $\pm$ 0.057
Ours	0.860$\pm$0.086	0.855$\pm$0.047	0.923$\pm$0.069	0.879$\pm$0.051
MS-CMRSeg				
FullSup	0.883 $\pm$ 0.058	0.856 $\pm$ 0.042	0.936 $\pm$ 0.034	0.892 $\pm$ 0.035
pCE	0.694 $\pm$ 0.076	0.620 $\pm$ 0.077	0.797 $\pm$ 0.078	0.704 $\pm$ 0.049
W/O CT	0.710 $\pm$ 0.085	0.626 $\pm$ 0.075	0.793 $\pm$ 0.067	0.710 $\pm$ 0.048
Ours	0.727$\pm$0.083	0.652$\pm$0.078	0.826$\pm$0.059	0.735$\pm$0.046

TABLE VII

EFFECTIVENESS EVALUATION OF MOMENT-UPDATE STRATEGY FOR SEMANTIC-LEVEL CLASS-TOKENS

Method	DSC $\uparrow$			
	RV	myo	LV	mean
FullSup	0.882 $\pm$ 0.095	0.883 $\pm$ 0.042	0.930 $\pm$ 0.074	0.898 $\pm$ 0.070
pCE	0.625 $\pm$ 0.160	0.668 $\pm$ 0.095	0.766 $\pm$ 0.156	0.686 $\pm$ 0.137
Queue-update	0.854 $\pm$ 0.090	0.849 $\pm$ 0.049	0.916 $\pm$ 0.078	0.874 $\pm$ 0.054
Moment-update	0.860$\pm$0.086	0.855$\pm$0.047	0.923$\pm$0.069	0.879$\pm$0.051

C. The Impact of Semantic-Level Class-Tokens

We examined the effectiveness of semantic-level class-tokens, similar to pixel-level queries. Results (Tab. VI) demonstrate that the inclusion of class-tokens enhances segmentation in both the ACDC and MS-CMRSeg Challenge datasets.

The addition of semantic-level class-tokens improved segmentation performance on the ACDC dataset from 87.2% to 87.9% and on the MS-CMRSeg Challenge dataset from 71.0% to 73.5%. These tokens contribute to the model's understanding of context and class-specific knowledge, resulting in improved accuracy.

D. The Impact of Moment-Update Strategy for Semantic-Level Class-Tokens

In our exploration of storing semantic-level class-tokens, we experimented with two methods: employing a queue for multiple tokens or maintaining a single token per class. Initially, we adopted a queue-based approach, similar to the pixel-level queries method. However, as indicated in Tab. VII, this resulted in a performance decline when storing numerous semantic-level class-tokens, reducing segmentation accuracy from 87.9% to 87.4%. This decrease can be attributed to the sparse nature of scribble annotations, which provide limited pixel-level annotations per image. Consequently, this inconsistency in guidance emerged as different class-tokens were generated for the same class.

To overcome this challenge, we transitioned to using a single token per class for all images in the dataset. This modification improved the representation of each class, addressing the issue of inconsistent guidance. We then employed separate storage and update methods for pixel-level queries and class-tokens.

E. The Impact of Feature Similarity Matching Using Hard and Soft Similarity

In the context of feature similarity comparison, we employ the Softmax function to calculate soft similarity between

TABLE VIII

EFFECTIVENESS EVALUATION OF HARD AND SOFT SIMILARITY

Method	DSC $\uparrow$			
	RV	myo	LV	mean
FullSup	0.882 $\pm$ 0.095	0.883 $\pm$ 0.042	0.930 $\pm$ 0.074	0.898 $\pm$ 0.070
pCE	0.625 $\pm$ 0.160	0.668 $\pm$ 0.095	0.766 $\pm$ 0.156	0.686 $\pm$ 0.137
Pixel Hard - Sematic Hard	0.851 $\pm$ 0.096	0.851 $\pm$ 0.049	0.915 $\pm$ 0.084	0.872 $\pm$ 0.058
Pixel Hard - Sematic Soft	0.850 $\pm$ 0.099	0.851 $\pm$ 0.049	0.914 $\pm$ 0.079	0.872 $\pm$ 0.058
Pixel Soft - Sematic Hard	0.852 $\pm$ 0.099	0.847 $\pm$ 0.054	0.912 $\pm$ 0.090	0.870 $\pm$ 0.064
Pixel Soft - Sematic Soft	0.860$\pm$0.086	0.855$\pm$0.047	0.923$\pm$0.069	0.879$\pm$0.051

TABLE IX

EVALUATION OF NUMBER OF PIXELS PER CLASS

Method	DSC $\uparrow$			
	RV	myo	LV	mean
500	0.851 $\pm$ 0.094	0.847 $\pm$ 0.056	0.913 $\pm$ 0.092	0.870 $\pm$ 0.063
1000	0.860$\pm$0.086	0.855$\pm$0.047	0.923$\pm$0.069	0.879$\pm$0.051
2000	0.854 $\pm$ 0.095	0.853 $\pm$ 0.047	0.911 $\pm$ 0.085	0.873 $\pm$ 0.057
5000	0.851 $\pm$ 0.099	0.849 $\pm$ 0.053	0.915 $\pm$ 0.090	0.872 $\pm$ 0.062

training pixel-level features and those in the queries and class-tokens. To assess its effectiveness, we introduced a binary representation (ranging from 0 to 1) for selecting the most similar feature.

Tab. VIII presents results using two approaches: hard similarity (binary) and soft similarity (Softmax). Soft similarity outperforms hard similarity due to its gradual transition in the comparison process, making it more robust to noise. Soft similarity utilizes probabilities, allowing for a nuanced understanding of feature similarity rather than relying on a strict binary choice.

Therefore, our findings suggest that using soft similarity is more effective for feature matching. It offers a flexible and probabilistic way to measure similarity, resulting in better segmentation performance.

F. The Impact of Queue Size

In our current approach, we use queries with 1000 pixels for each class. To determine the optimal number, we conducted experiments with different sizes (500, 1000, 2000, and 5000 pixels per class). As indicated by the results in Tab. IX, using 1000 pixels per class yields the best performance. This size strikes a balance by providing sufficient information for each class without introducing excessive computational overhead. When considering the cases with more or fewer pixels, the observed decline in performance might be attributed to the potential confusion introduced during the comparison process. The excess or scarcity of similar pixels compared to the target pixels could lead to a certain level of ambiguity. In contrast, 1000 pixels appear to be an appropriate size for storing pixels in the ACDC scribble dataset. Therefore, we adopt a query size of 1000 pixels per class.

VI. VISUALIZATION

In Fig. 8, we showcase the segmentation results of our model on the ACDC dataset, comparing it with state-of-the-art scribble-based learning methods. The visual comparison highlights our model's proficiency in boundary identification, revealing more accurate and precise segmentation boundaries compared to other methods. Our approach successfully tackles the challenge of limited supervision posed by scribbles, preventing significant misclassification of large regions.

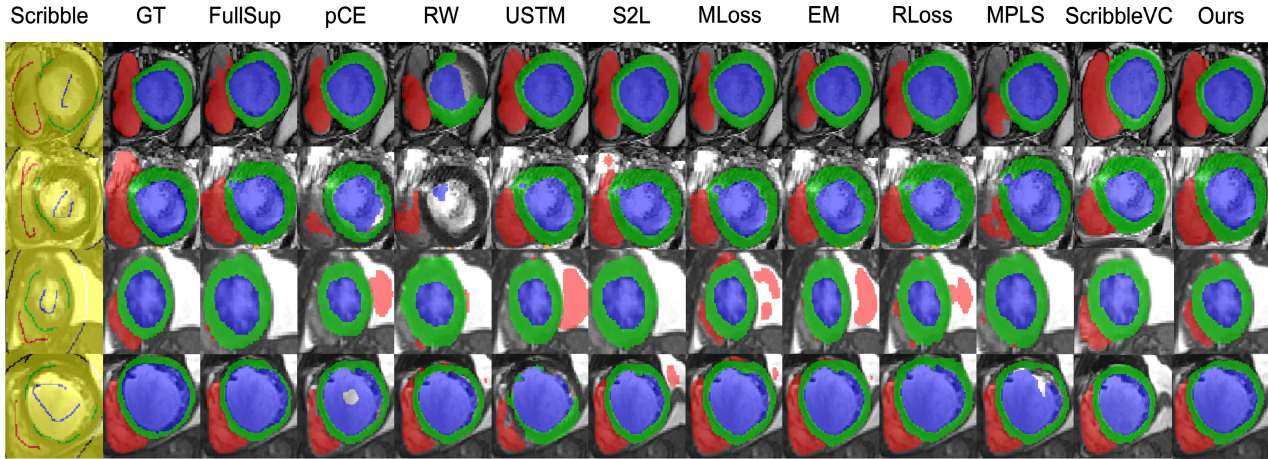


Fig. 8. Segmentation results on ACDC dataset using different approaches. Colors represent RV, myo, and LV classes (red, green, and blue).

The integration of our matching strategy, which utilizes pixel-level queries and semantic-level class-tokens as references, effectively addresses inconsistencies introduced by different scribble types, contributing to enhanced segmentation accuracy for medical images.

VII. CONCLUSION AND DISCUSSION

In conclusion, weakly-supervised learning with scribble annotation holds promise for reducing medical image segmentation costs. However, its performance is constrained by challenges such as incompleteness and subjectivity. To address these limitations, we introduce a novel matching-based approach that capitalizes on diverse pixel-level features from various scribble types to construct a comprehensive reference set. This set enhances the network's understanding of the target region by learning consistent patterns. Our experiments on three datasets showcase significant improvements in segmentation accuracy. Future research directions include exploring alternative annotation types, refining the construction of the reference set, and adapting the approach to diverse medical imaging modalities and tasks, such as tumor and organ segmentation.

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